ICICT 2026
International Congress & Excellence Awards

Unified Health Index as a Service for Cross-Jurisdiction Emergency Care

Authors

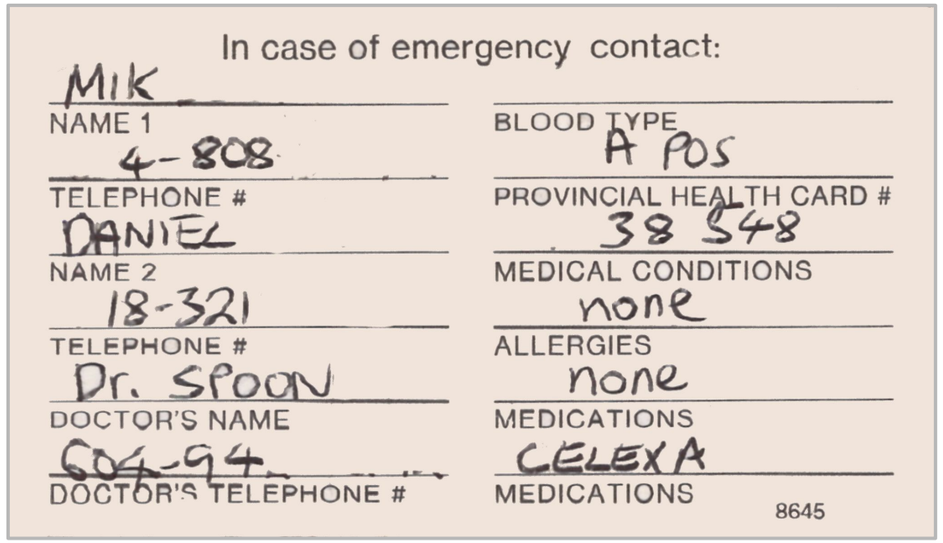
Fernando Latorre [1 3]
Iván Becerro [2]
Nuria Sala [1 3]

Affiliations

[1] Conéctate Soluciones y Aplicaciones, S.L., Soria, Spain
[2] Accuro Technology S.L., Madrid, Spain
[3] Fundación UNID, Soria, Spain

connect health
accuro technology
UNID fundación

The Traditional Model



This is what emergency identity has looked like for decades:

A physical card with personal, static data.

- Name
- Blood type
- Allergies
- Emergency contacts
- Medication

- No dynamic updates
- No section-level access control
- No cross-border interoperability

Unified Health ID for Emergencies and Care Continuity

Instead of printing the data, you print a QR code and a *did:web* identifier

Unified Health Identity

did:web:connector.example.com:
individual:multibase:z012345abcde543210edcba

Now the card stores an **identity pointer**, **not personal and medical data**.

- The did:web is resolved as a regular HTTPS web link.
- It returns a **DID Document**, which **contains:**
  - **Identity verification** (public signature keys)
  - **Where to recover data** (service endpoints)
  - **Digital identities of legal representatives.**

You can also display the *did:web* on an SOS screen (Android, iOS, etc.)

Enabling the Unified Health Index (UHIx)

Data is linked, but not moved

The Unified Health Index (UHIx) is a new service on top of current FHIR local data providers.

Providers notify UHIx of new record links under the individual's authorization.

The individual does not need to manually retrieve and index records through a web or mobile app.

Consent → Index + Policies → Secure Accessibility

Unified Health Index as a Service

How We Make It Possible

Imagine that a person's health history is a large book, but **each chapter** (tests, reports, procedures) **is stored in a different library** (hospitals, clinics).

Today, to read the full book, you have to travel to each library, request permission, and compose your index.

The Unified Health Index does not copy the chapters of your book.
It receives updates to store the links with the information of which library and which shelf each chapter is on.

1. The organization connects to the data space (network)

2. The patient grants permission

3. The request is resolved instantly

Unified Health Index for the International Patient Summary (IPS)

Activation through **signed consent.**

Free choice of **provider.**

Links updated in **real time.**

Inclusivity and accessibility for all.

Data for statistics and digital twins.

Permissions based on role (actor type) and:

- **region**,
- **organization**, or
- **e-mail address**.

Three-Layer Model

👤 Identity Layer

- **UHI = *did:web*** (decentralized, unified health identity)
- **Verifiable credentials**
- **Public keys**

No clinical data in the digital identity.

📚 Index Layer

- **UHIx:** externalized index service for the International Patient Summary (IPS).
- **Policies:** permit/deny
- **Granular access tokens:** SMART-On-FHIR

No clinical data on blockchain.

🗄️ Data Layer

- **Patient portals:** authoritative clinical records.
- **DICOM**: medical imaging repositories.
- **Device APIs:** telemetry from wearables and medical sensors.

Decentralized system. No central repository.

Emergency Access and Break-Glass Logic

🟢 Tier 1: Pre-Authorized Emergency Access

- **Role + jurisdiction match**
- **Scoped SMART token**
- **IPS sections retrieved**
- **Fully policy-driven**

Standard emergency access.

🟡 Tier 2: Curated Emergency Dataset

- **Individual-defined** emergency summary: contacts, allergies, implants, etc.
- **Quick access**
- **Minimal necessary scope:** doctor, nurse, paramedic.

Predefined emergency dataset.

🔴 Tier 3: Break-Glass Escalation

- **Legal emergency doctrine** (vital interest)
- **Verified doctor**
- **Signed justification**
- **Time-limited** expanded scope
- **Full audit trail**
- **Notification** after event

No silent override. Always auditable.

Clinical custodians must validate scoped access tokens before disclosing records

Data Anonymization

What Are Metadata?

They are data sets that do not allow identification of the originating individual:

- Sex
- Year of birth
- Jurisdiction where the data is created
- Date of data creation
- Clinical or product code

What is a digital twin?

They are metadata of an unknown individual over time (their history).

- Health status at a specific point in time (“snapshot” of the digital twin).
- Evolution over time.
- Forecast of possible pathways.

What Is Anonymized and When?

- **Metadata when certifying a document.**
- **Digital twin:** updated by the index provider if there is no revocation of consent.

Data Space: Trust Architecture

1. Governance Entity (Root CA)

- Signs the credentials of intermediate certification entities (ICAs)..
- Issues credentials to intermediate verification entities.
- Oversees software and algorithm ethics.

2. Intermediate Certification Entity (ICA)

- Verifies the authenticity of organizational information.
- Issues and signs organizational credentials.

3. Node Operator (host)

- Provides infrastructure for service providers.
- Validates digital identity evidence submitted by organizations during the registration process.

4. Service provider (API connector)

- Registers by enrolling with an Operator.
- Reserves activation licenses for employees and end users (provided by the Operator).
- Issues employee credentials.

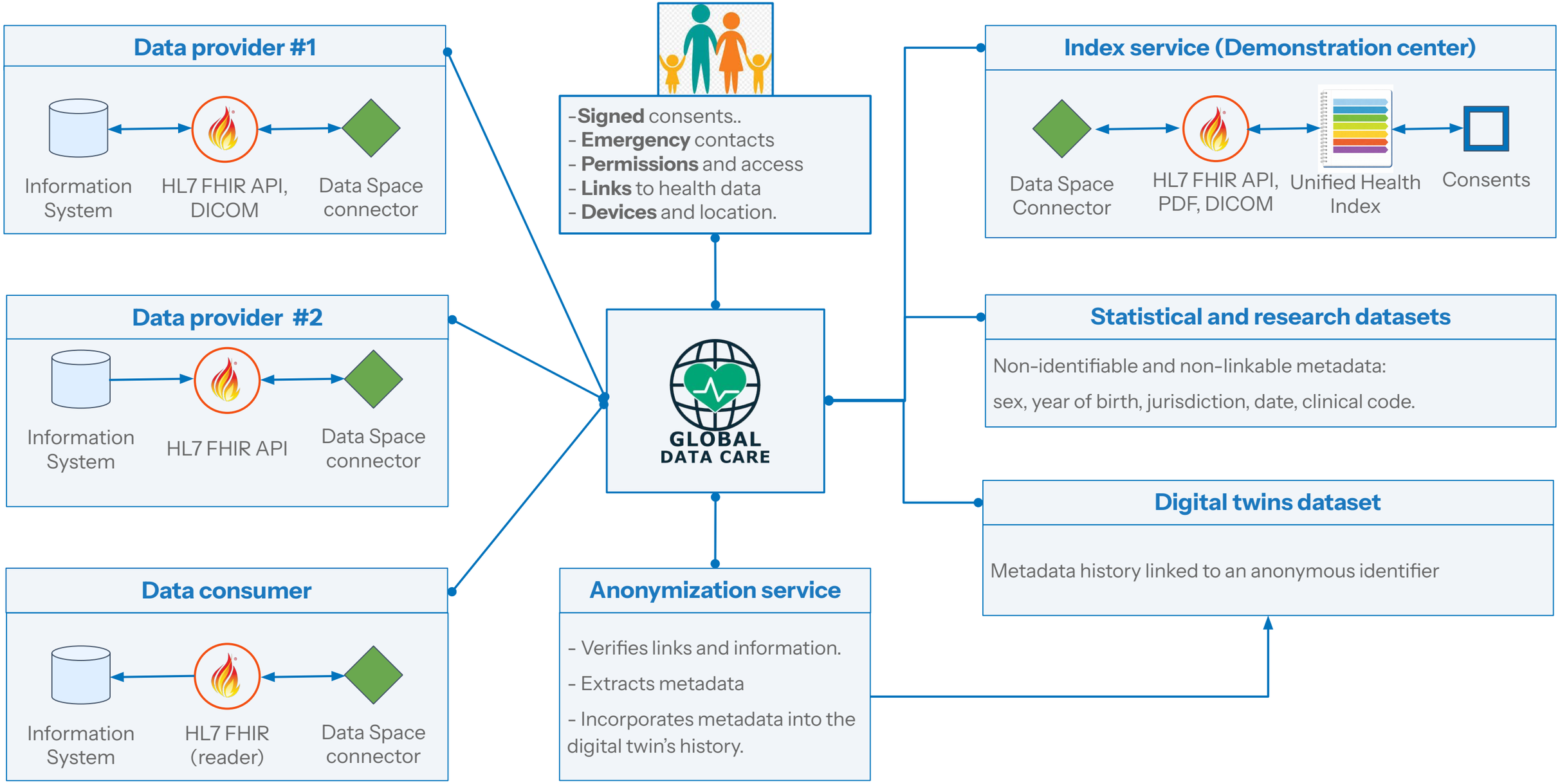
5. Employee (professional role)

- Linking cryptographic keys (digital identity) to their email and employee role.
- Verifies and registers end-user identity evidence.
- Registers, requests, and reads data: consents, index, documents, related persons, etc.

6. End user (individual)

- Signs the consent to activate the index.
- Registers permissions, health data, and related persons.

Data Flow and Sharing Scheme



Final Words

Emergency data access must be cross-border, fast, and auditable:

- 👤 **UHI enables discovery**: where the Unified Health Index (UHIx) resides.
- 📚 **UHIx enables controlled access** to authorized data links.
- 🗄️ **Clinical data remains with original custodians**.
- 🚨 **Emergency access is scoped, traceable, and governed**.

Thank you for attending!

Fernando Latorre
ORCID: 0009-0004-6313-6471
Email: ferlatorre78@gmail.com

ICICT 2026
International Congress & Excellence Awards

More info: accuro.es
Docs: github.com/global-datacare/docs